

## **Machine Learning Based Food Waste prediction**

**ID:** 22201855

**ID:** 24141121

**Abstract:** Many kitchen whether it is commercial or institutional they face problems many challenges due to overproduction of food. This create a load of food waste. This study develops a ML supervised model like Linear Regression, Decision Trees and Neural Network to predict food\_waste\_kg based on features like kitchen\_staff, staff\_experience, meal\_served, temperature\_C, humidity\_percent, waste\_category, special\_event and also past\_waste\_kg. We preprocessed through the pipeline to handle missing values, scale numerical features and encode categorical data. Cross Validation shows us that the models has achieved the better r2 when they use crossValidation . Later we also divided the dataset into 3 clusters using Kmeans.

**Introduction:** This project aims to predict continuous values using Machine Learning models for daily kitchen\_waste\_kg using others feature of the dataset like kitchen\_staff, staff\_experience, meal\_served, temperature\_C, humidity\_percent, waste\_category, special\_event and also past\_waste\_kg. Later we will also use Kmeans to divide the dataset into clusters.

**Description:** The dataset has 911 rows and 12 columns including the target column (food\_waste\_kg). The target column has continuous values which makes this problem a regression problem. The dataset has both categorical and numerical columns. It has 8 numerical columns and 3 categorical columns. And the target column is numerical. We need to encode the categorical columns because Machine Learning models will not be able to predict without numerical values if we leave them as a categorical variable the model won't be able to predict. We can see from the correlation matrix that food\_waste\_kg is strongly correlated with meals\_served and past\_waste\_kg. But columns like temperature\_C, humidity\_percent and days\_of\_week have negligible correlation with the target column. As the target column has regression values it does not have unique classes.

### **Dataset Preprocessing:**

- **Missing/Null Values:** The staff\_experience contains 164 nan values while others have no null values. We used imputer which will impute most\_frequent values will get imputed and replace the nan values
- **Categorical Columns:** The dataset has categorical values which need to be encoded for the machine learning model's prediction. Applied OneHotEncoder(handle\_unknown='ignore', sparse\_output=False) within a ColumnTransformer pipeline which will encode based on class.

- **Feature Scaling:** Numerical features which have big differences are needed to be scaled. Otherwise the model will get biased to them. We will use standard scaler in pipeline to scale them. It will scale the values to in between 0 to 1.

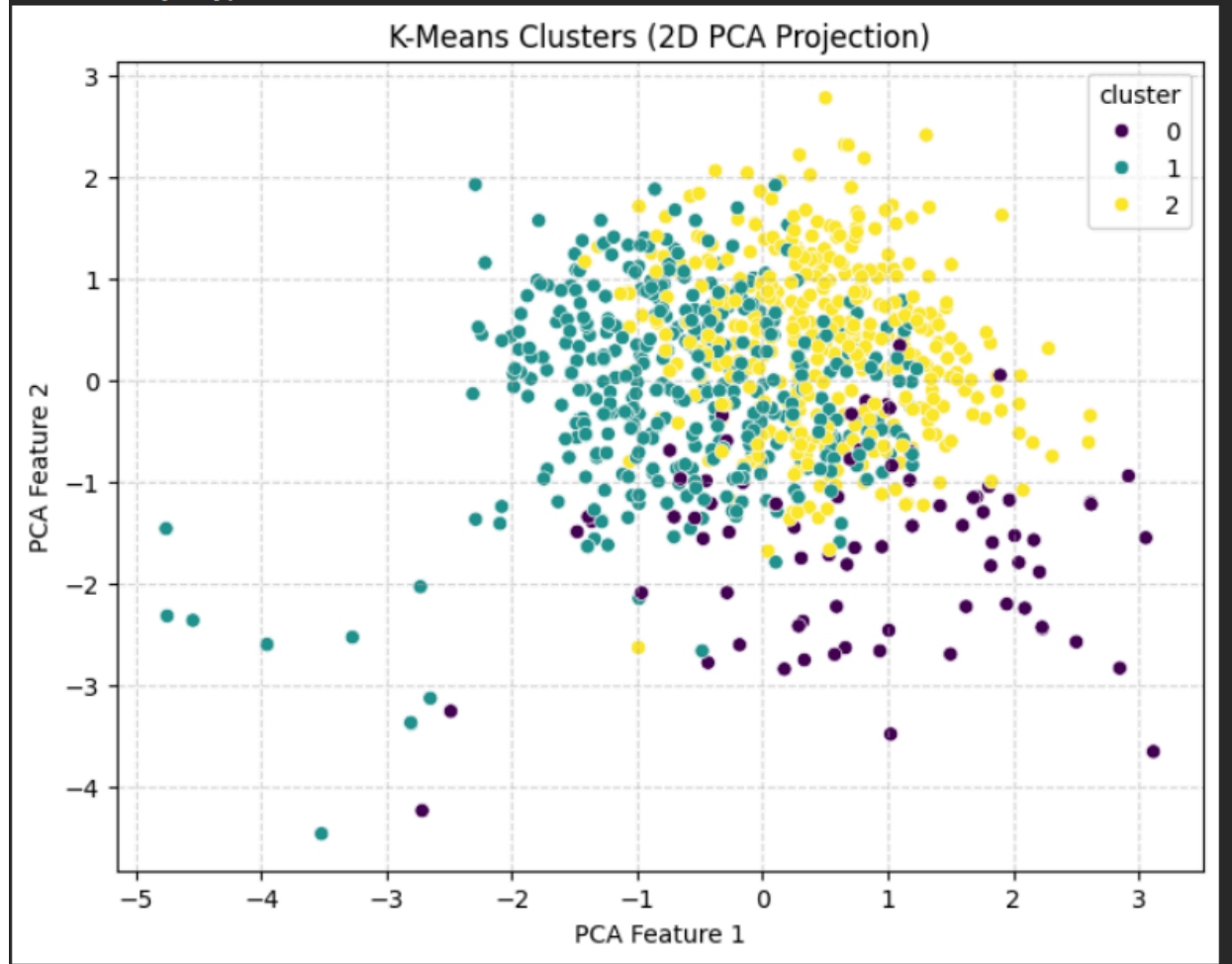
**Dataset Splitting:** We split the dataset using `train_test_split` 80% (728 samples) for the training set and 20%(183 samples) for the test set. We used k fold Validation (`KFold(n_splits=5, shuffle=True, random_state=42)`) is implemented on the training data. This ensures every fold acts as a validation set once during training evaluation.

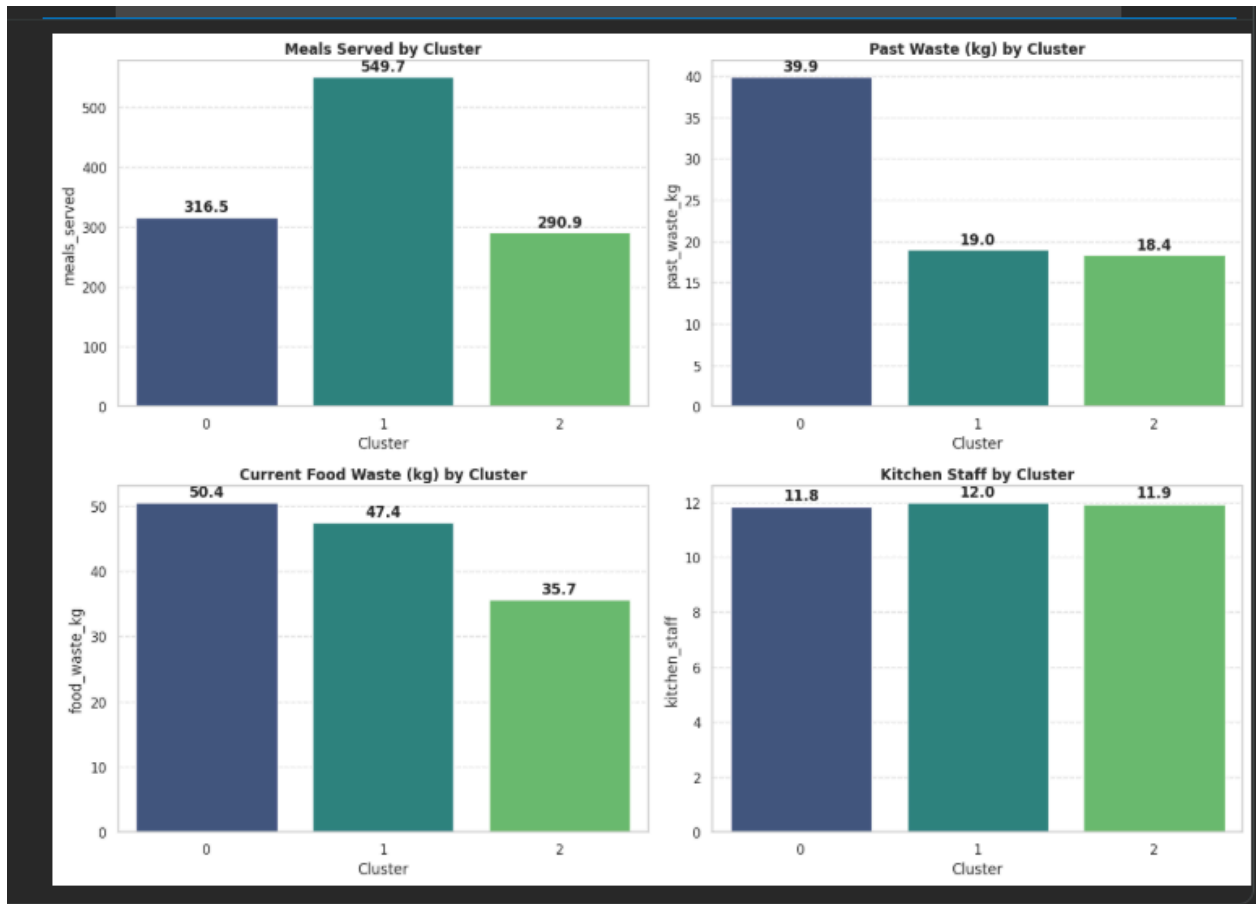
### **Model Training and Testing:**

- **Linear Regression:** Linear Regression fits a linear equation to the features to predict the continuous output `food_waste_kg`. We used 5-fold cross-validation on the training set. It estimates future or unknown continuous values based on existing data trends.

- **DecisionTree Regression:** The Decision Tree Regressor splits the dataset to predict food\_waste\_kg. It solves problems by asking a series of yes-or-no questions that lead to a final choice or numeric score.
- **Neural Network:** The model uses mlp regressor multiple perceptrons and hidden layer to predict food\_waste\_kg. It helps the model to recognize the model and give it's prediction.
- **Kmeans:** We used this model to transform dataset into clusters. This is a unsupervised model. This model fits our data and transform our data into 3 cluster.

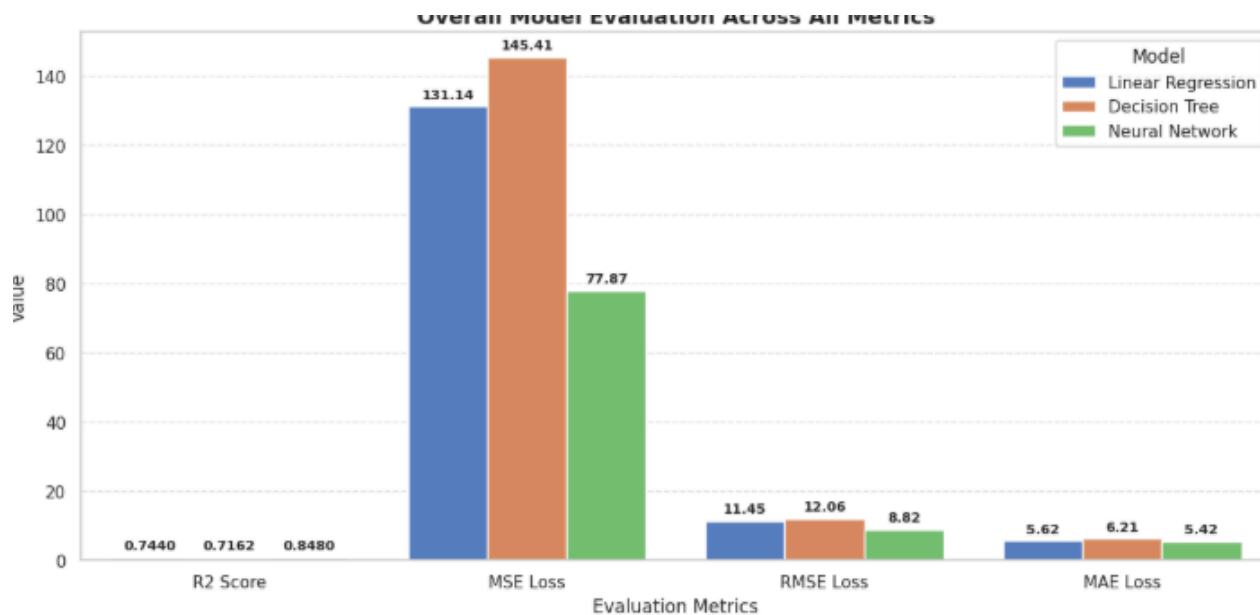
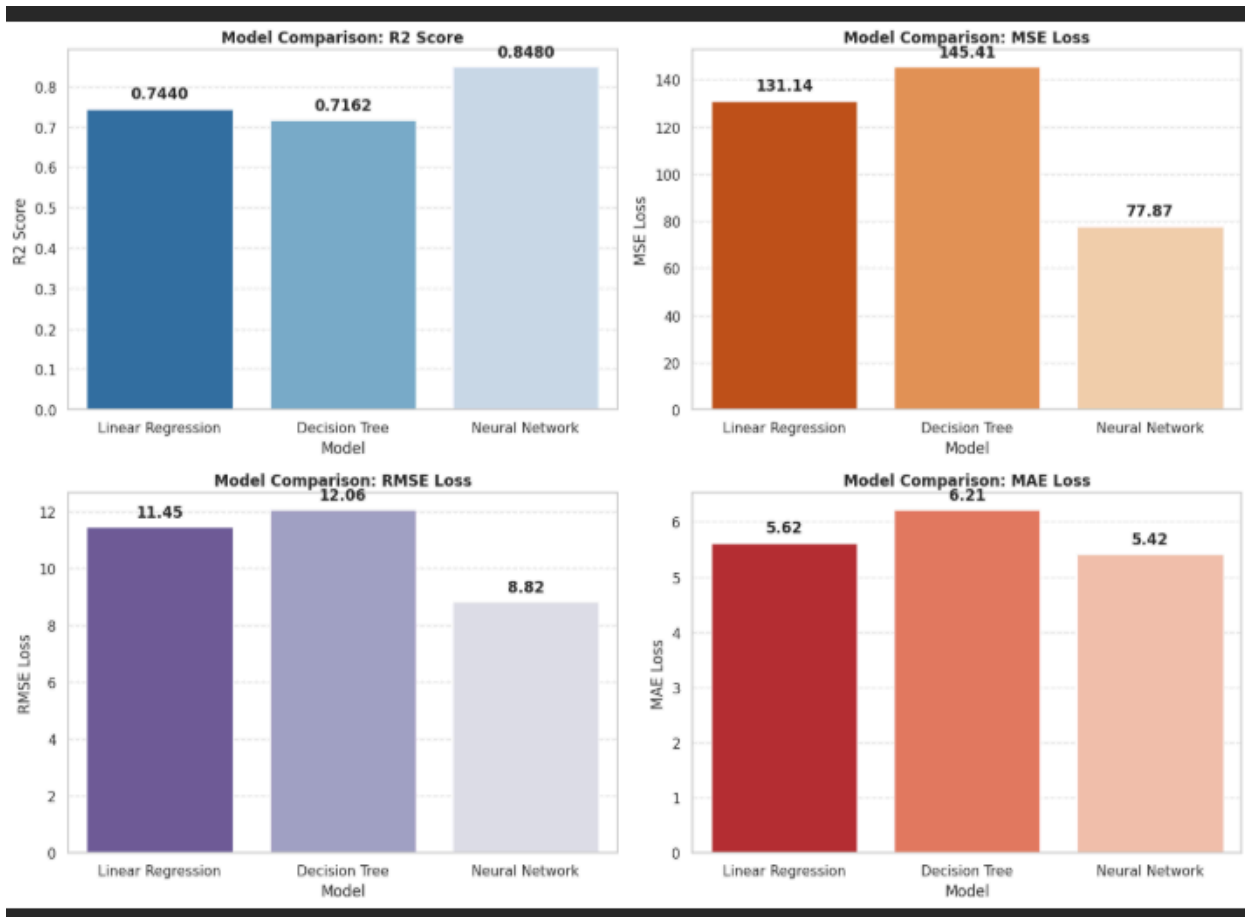
```
Samples per cluster:  
cluster  
1    422  
2    411  
0     78  
Name: count, dtype: int64
```





## Model Selection & Performance Comparison :

|   | Model             | R2 Score | MSE Loss   | RMSE Loss | MAE Loss |
|---|-------------------|----------|------------|-----------|----------|
| 0 | Linear Regression | 0.744044 | 131.143728 | 11.451800 | 5.619293 |
| 1 | Decision Tree     | 0.716194 | 145.413133 | 12.058737 | 6.213790 |
| 2 | Neural Network    | 0.848024 | 77.867620  | 8.824263  | 5.415816 |



Based on all Evaluation Metrics, The Neural network model is the best model overall for highest  $r^2$  score and lowest mse,rmse,mae loss. Linear regression has second best  $r^2$  score and second lowest mse,rmse,mae loss. The Decision tree model performed worst among the three models. Its  $r^2$  score is the lowest and loss are the highest among all 3 models.

**Conclusion:** The neural network model performed better because it has multi layer hidden structure which helps to identify pattern so that it performed better. Linear regression comes second with it's prediction and decision tree comes third. Kmeans clustering clustered our dataset into 3 clusters. I faced challenge in this project as it has missing values, high value difference and overfitting issues.